ASSIGNMENT REPORT- 6

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For The Subject

Pattern Recognition and Machine Learning

(26AI5707)



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OCTOBER 2026

INTRODUCTION:

This assignment observes the generation of 2D data using a Gaussian Mixture Model (GMM) and then applying K-means clustering and classification based on distortion. Each of the two classes is modeled using a mixture of three Gaussian components. The data are generated from three different covariance settings: spherical, diagonal and full covariance. The generated points are visualized, divided in training and testing sets in 80:20 ratio and used to study the clustering boundaries and classification performance.

METHODOLOGY:

1. Gaussian Mixture Model

It represents a probability density as a weighted sum of Gaussian component densities. For a 2D observation x, the model is:

$$p(x) = \Sigma_{i=1}^{K} \pi_i \, N(x; \mu_i, C_i)$$

where,

K - number of components

π_i - mixture weight of component i

μ_i - mean vector

C_i - covariance matrix

2. Box–Muller Transformation

It is used to generate two independent standard normal random values from two uniform random values u_1 and u_2.

$$Z_1 = \sqrt{(-2 \ln U_1)} \cos(2\pi U_2)$$
$$Z_2 = \sqrt{(-2 \ln U_1)} \sin(2\pi U_2)$$

3. Gaussian Point Generation

A standard normal pair is transformed using the mean and covariance of the selected component. For a symmetric positive-definite covariance matrix C = [[a, b], [b, c]], the point is generated as:

$$x_1 = \mu_1 + \sqrt{a}\, z_1$$
$$x_2 = \mu_2 + (b/\sqrt{a})z_1 + \sqrt{(c - b^2/a)}z_2$$

This transformation handles the spherical, diagonal, and full covariance matrices

4. Training and Testing Split

The generated data of each class is shuffled and 80% is used for training data and 20% is used for test data. we generate 1000 samples per class, such that each class has 800 training points and 200 testing points for each covariance configuration.

5. K-Means Clustering

Three initial centroids are selected using an initial farthest point initialization procedure. Each training point is assigned to the nearest centroid in terms of squared Euclidean distance. Then each centroid is moved to the mean of the points assigned to it. The steps are repeated until the centroids move by less than a small tolerance, or the maximum number of iterations is reached. K-means models are fitted separately for the two classes.

$$d^2(x, \mu) = (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2$$

6. Classification Using Distortion

For a test point x, distortion is defined as the minimum squared distance from x to any centroid of a class. The point is assigned to the class with the minimum distortion:

$$D_k(x) = \min_j \|x - \mu_{kj}\|^2$$

If $D_1(x) < D_2(x)$, the point is classified as Class 1; or else, it is classified as Class 2. The held-out 20% test data is used to calculate accuracy and a confusion matrix.

RESULTS AND OUTPUT:

Spherical
Class 1 - Training: 800 Testing: 200
Class 2 - Training: 800 Testing: 200

Diagonal
Class 1 - Training: 800 Testing: 200
Class 2 - Training: 800 Testing: 200

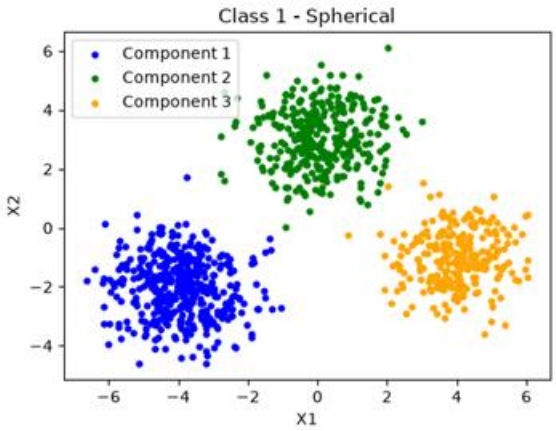
Full
Class 1 - Training: 800 Testing: 200
Class 2 - Training: 800 Testing: 200

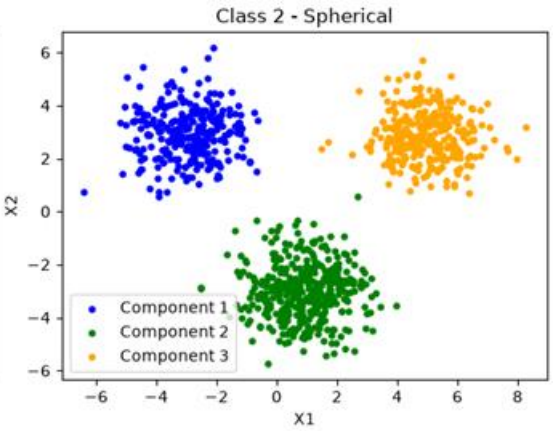
Covariance type	Class 1 test points	Class 2 test points	Accuracy
Spherical	200	200	96.75 %
Diagonal	200	200	94.5 %
Full	200	200	93.75 %

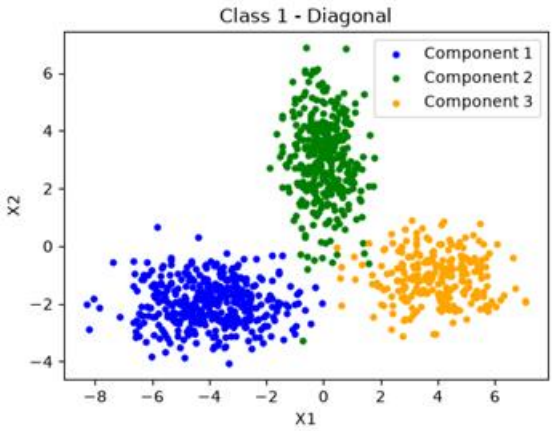
Spherical
Class 1 centroids:
[0.01, 3.019]
[-3.921, -1.958]
[4.016, -0.993]
Class 2 centroids:
[-2.941, 2.954]
[4.961, 2.873]
[0.867, -3.01]

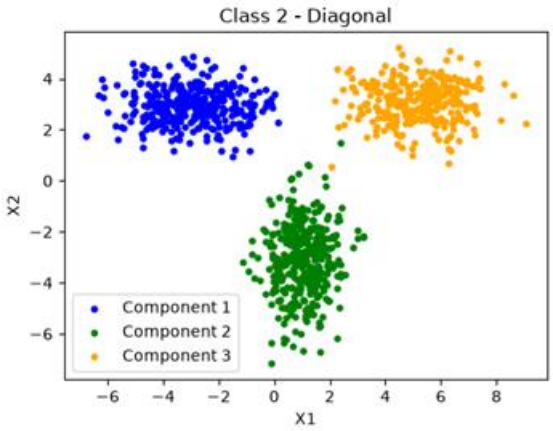
Diagonal
Class 1 centroids:
[3.895, -1.121]
[-3.933, -1.937]
[-0.037, 3.169]
Class 2 centroids:
[-2.995, 2.963]
[5.195, 3.104]
[1.031, -3.048]

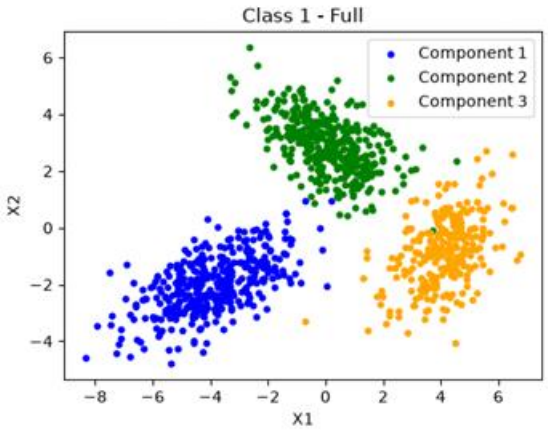
Full
Class 1 centroids:
[0.014, 2.84]
[-3.916, -2.004]
[4.083, -0.709]
Class 2 centroids:
[-2.981, 3.033]
[1.083, -3.036]
[4.923, 2.992]

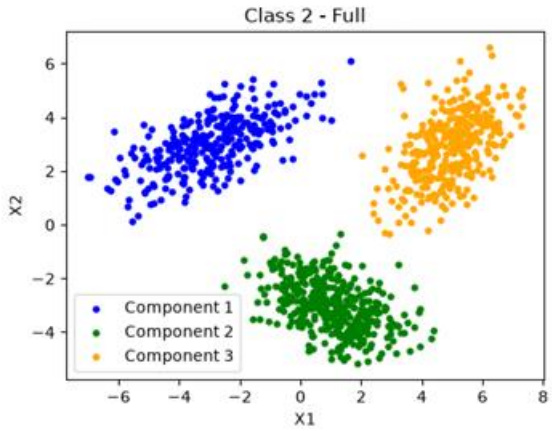






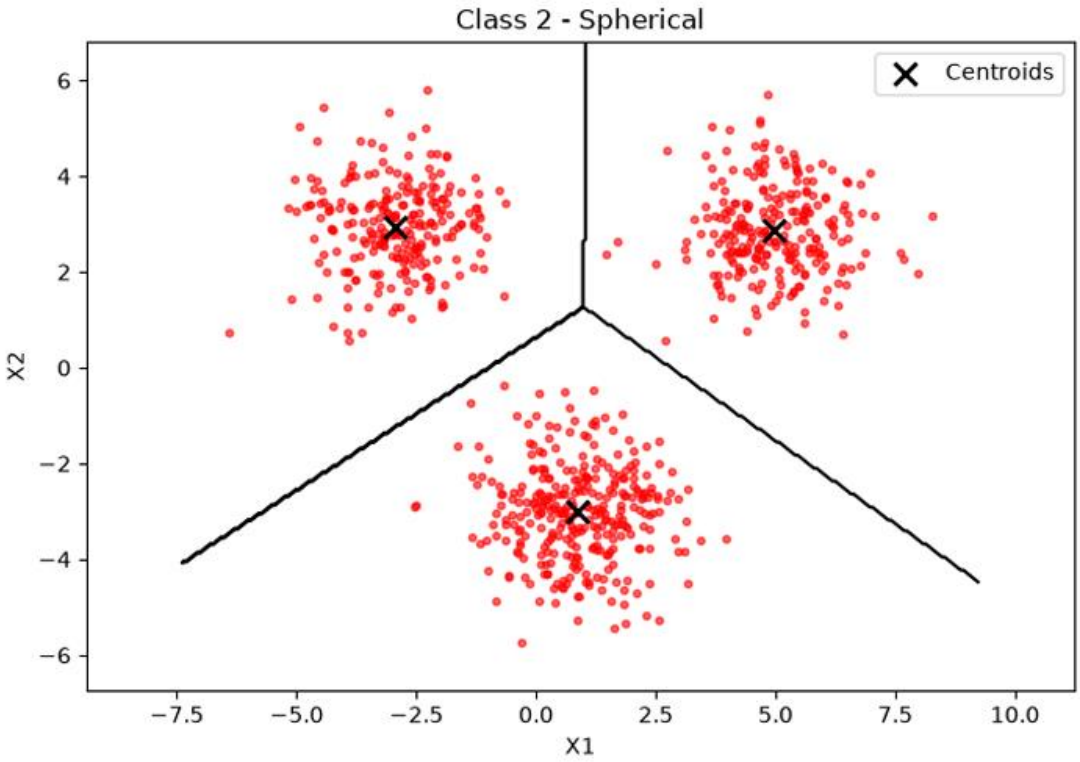


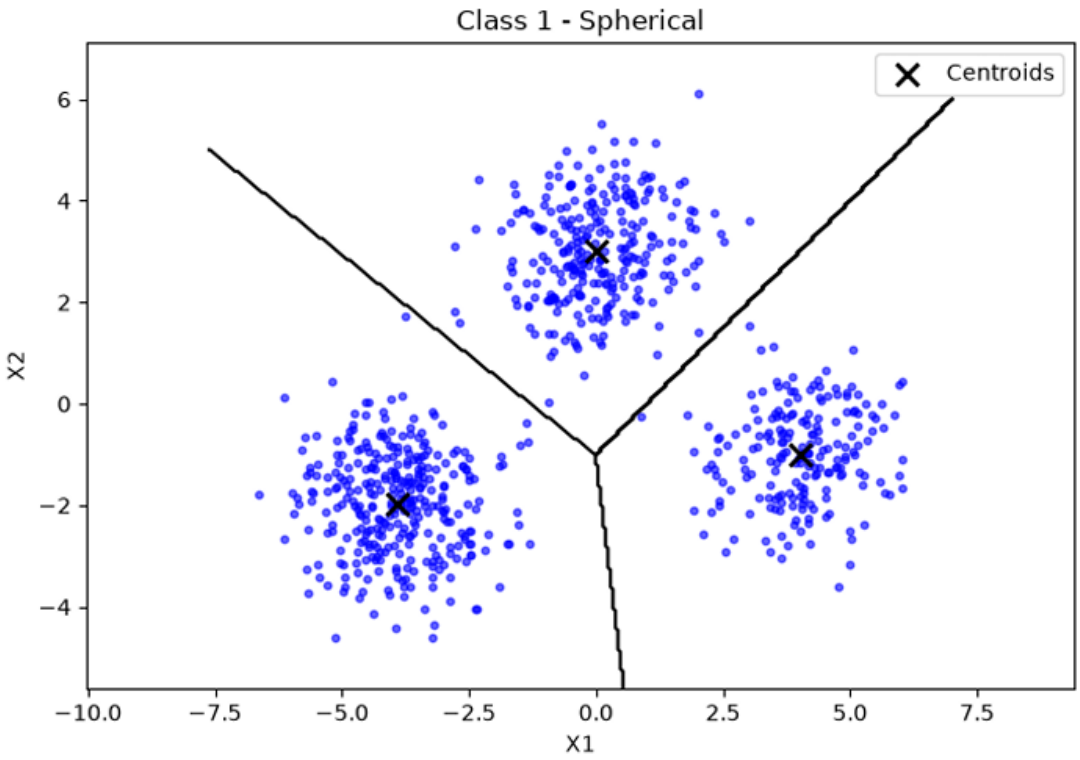




Covariance Type: Spherical
Test points: 400
Correct predictions: 387
Classification accuracy: 96.75 %
Confusion matrix:

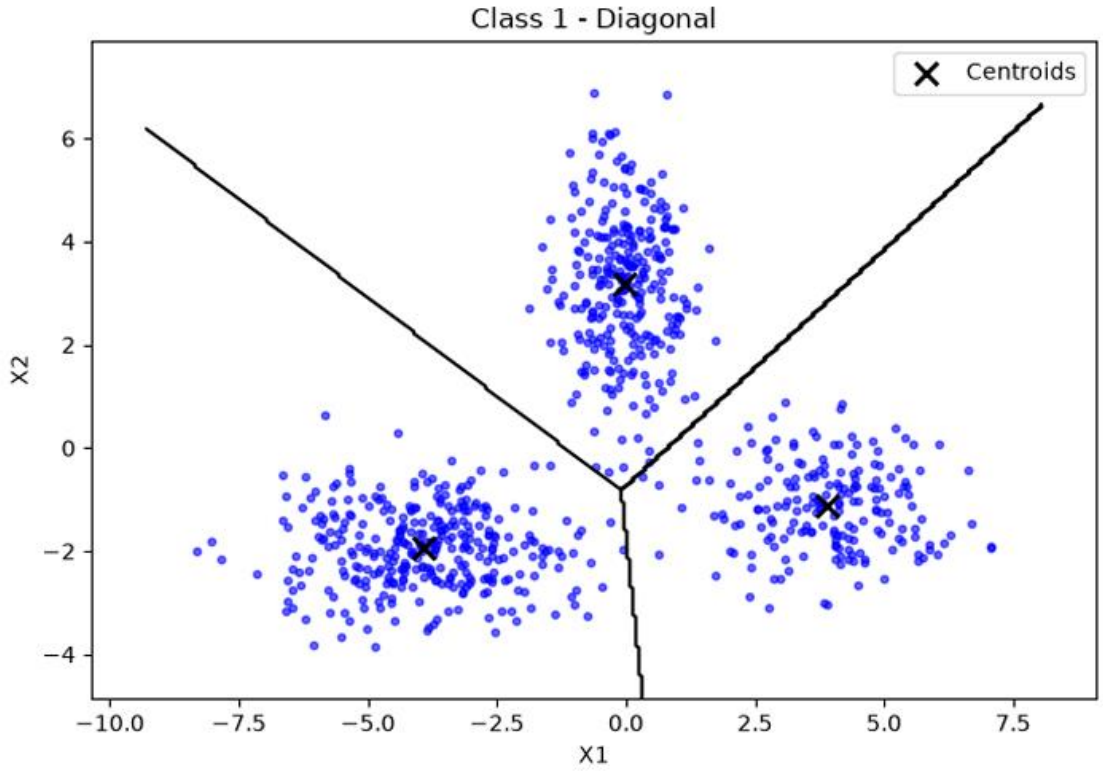
	Predicted Class 1	Predicted Class 2
Actual Class 1:	195	5
Actual Class 2:	8	192

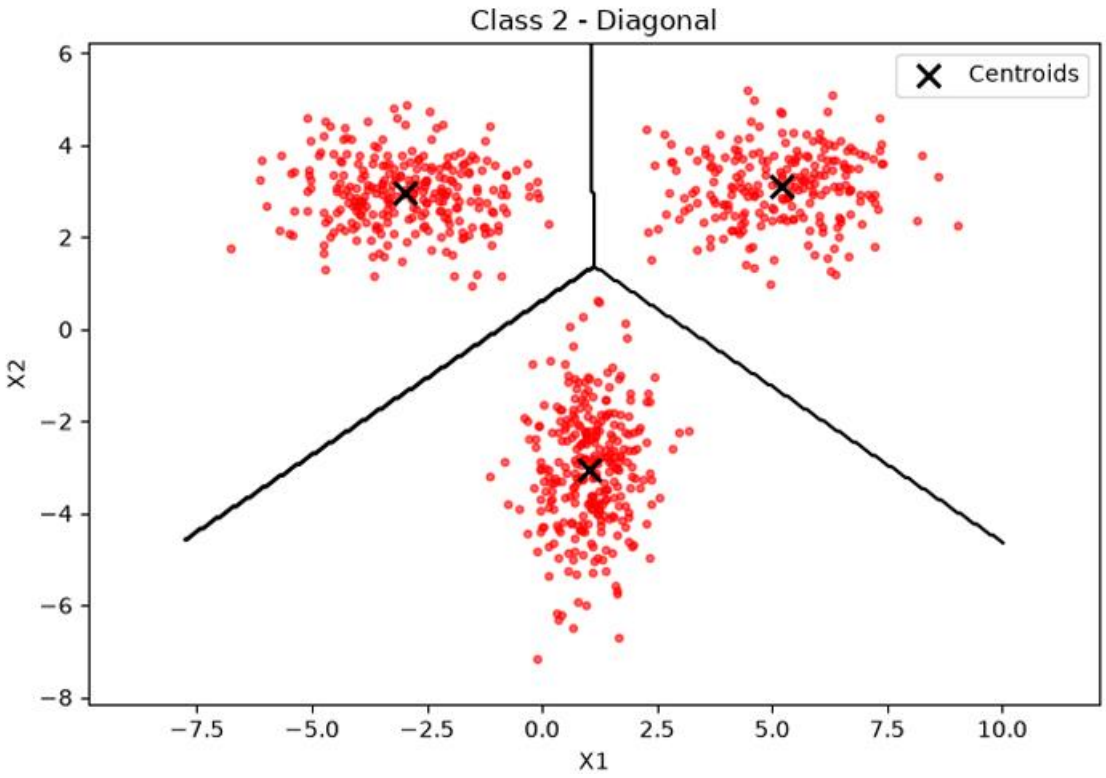




Covariance Type: Diagonal
Test points: 400
Correct predictions: 378
Classification accuracy: 94.5 %
Confusion matrix:

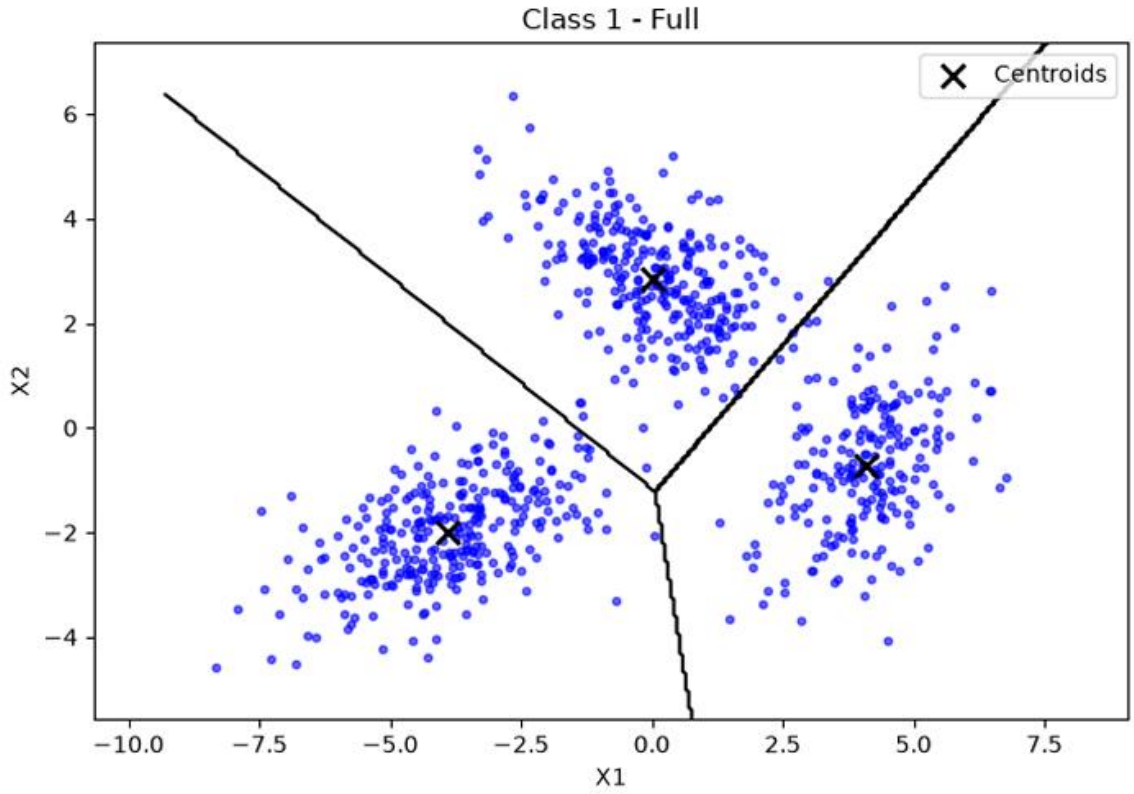
	Predicted Class 1	Predicted Class 2
Actual Class 1:	193	7
Actual Class 2:	15	185

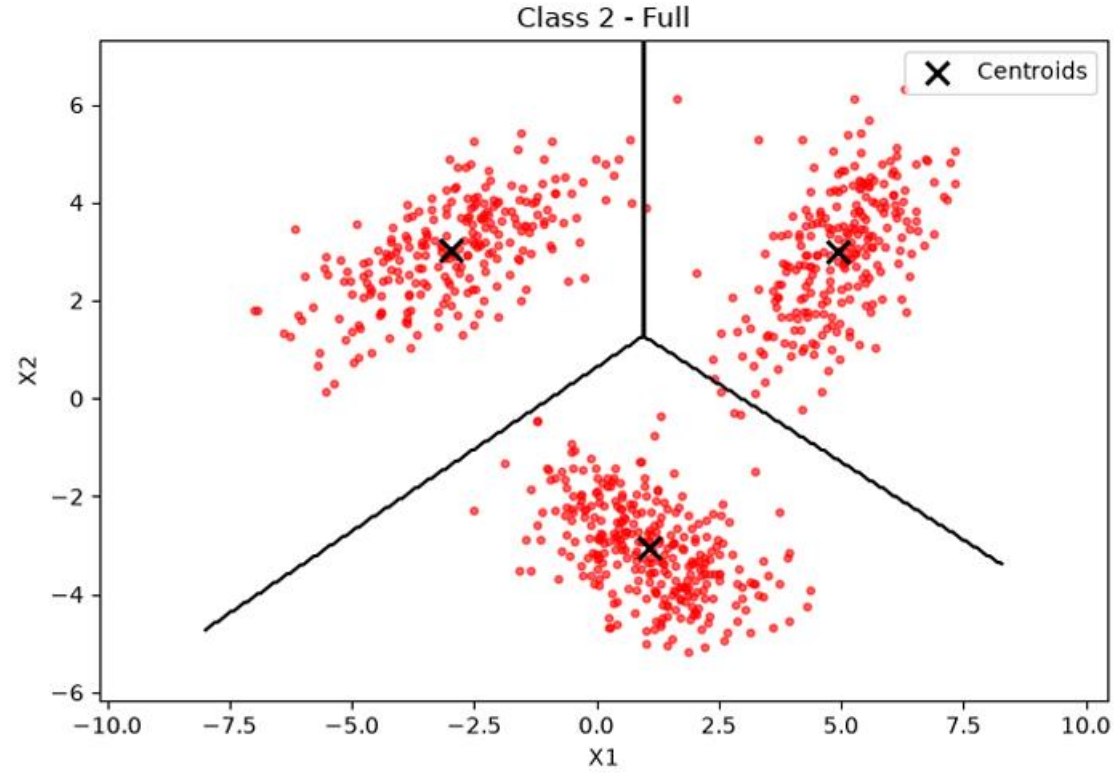




Covariance Type: Full
Test points: 400
Correct predictions: 375
Classification accuracy: 93.75 %
Confusion matrix:

	Predicted Class 1	Predicted Class 2
Actual Class 1:	187	13
Actual Class 2:	12	188

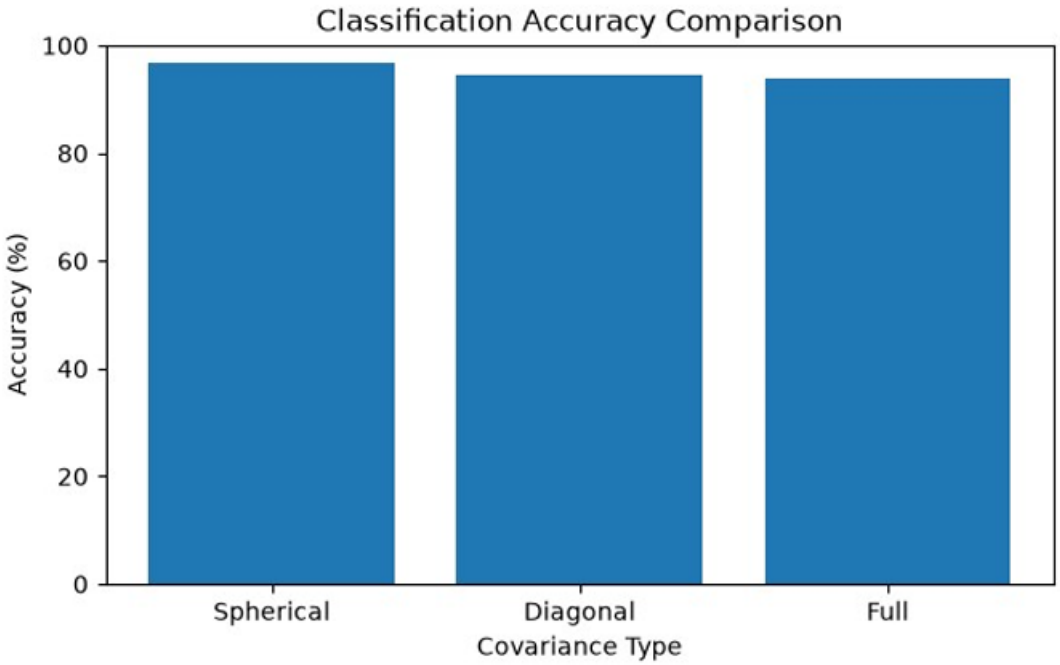




Covariance Type: Spherical
Class 1 mean training distortion: 1.9177
Class 2 mean training distortion: 1.9417

Covariance Type: Diagonal
Class 1 mean training distortion: 2.4261
Class 2 mean training distortion: 2.2564

Covariance Type: Full
Class 1 mean training distortion: 2.6827
Class 2 mean training distortion: 2.7801



Conclusion:

This assignment shows the implementation of generating 2D Gaussian mixture data and applying the manual K-means clustering for spherical, diagonal, and full covariance. Standard normal values are generated with the Box–Muller transformation and component selection is performed using mixture weights. The training data is used to learn class-specific centroids, while the test data is classified according to the minimum nearest-centroid distortion. The visualizations shows how covariance affects the data distribution and how K-means partitions the points.